

Creating Professional Circuit Designs Using NI Circuit Design Suite

Circuit design remains the main challenge to get an optimal performance for your controller board for any application. Here is how to approach your designs with professional tools and processes

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One of the greatest things about professional robots or other electronic gadgets that fascinates us is their customised circuit boards with no hanging wire or protruding part. Such designs are reliable and rugged and can outperform any controller board that is not designed for any particular application or made up on breadboards.

With the advent of PCB prototyping machine and its availability, manufacturing such a board has become quite inexpensive. Moreover, with widespread use of mobile phones, technical knowhow for fixing SMD components is easily available. You can get the chips placed with precision from any mobile repair shop. Thus circuit design remains the only challenge where you need to concentrate to get an optimal performance for your controller board for any application. Here we show you how to approach your designs with professional tools and processes.

NI Circuit Design Suite is a well known tool that electronics engineering students have been using for long. It is mostly used for circuit simulation, through National Instrument's Multisim package, which is a module in NI Circuit Design Suite. NI Ultiboard is the next module to Multisim in NI Circuit Design Suite. Using Ultiboard you can physically design a circuit board layout, design shape of the base board, place different components and do routing of different electrical networks. A 30-day trial package can be downloaded for free from the website of National Instruments. Engineering

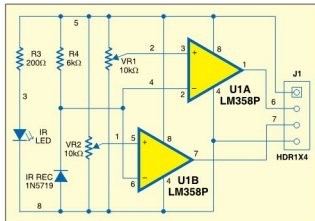


Fig. 1: Circuit for a line sensor

colleges may have its full version installed in their circuit simulation or design labs.

Typically, a circuit can be very easily constructed in Multisim environment and simulated to analyse different run-time parameters. Assuming that you know the circuit for any given application, you need to place the components on the board and route different networks. The best way I have always preferred to teach any subject is to start with an example and then practice with different sets. We will use Ultiboard, which is meant for the aforesaid purpose, with a commonly used IR line-sensing module. The circuit may look as shown in Fig. 1.

LM358 consists of two voltage comparators, U1A and U1B. Both of them are fed with two different reference voltages through potentiometers VR1 and VR2. The IR signal transmitted by the IR LED is sensed by a single IR-sensing module, and compared with two different threshold voltages through the two comparators. The out-

put through wires 6 and 7 is made available at the header. The header also supplies voltages for comparators and IR modules.

Normally, a line-sensing module utilises only one of the comparators for white-line detection. You may use both of them to work at different lighting conditions, without

putting much effort or additional components.

Designing a board is more of mechanical in nature than electronics. You need to know the footprints of all the devices you are going to use. This may require measuring the pin geometry using a Vernier calliper. Multisim and Ultiboard have very good library of various electronic devices and accessories, which includes their footprints.

The first step in circuit designing is to create a network of devices placed using wires, which is known as a 'netlist.' Two different approaches are followed for building the netlist for a given circuit, which is the primary requirement for designing a circuit. A simple approach is to create schematic on the Multisim workspace, by simply inserting the different components required for a circuit and connecting the leads through wires. This may be transferred to Ultiboard by pressing 'Transfer' tab in the main menu of Multisim. This creates a netlist file (*.ewnet). It will ask for default track